

Peer Skill Exchange & Mentorship Network

UML Component Diagram – Design Justification

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Course: Software Engineering

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1. Architectural Style

The Peer Skill Exchange & Mentorship Network follows a **Layered Architecture** consisting of the Presentation Layer, Business Layer, and Data Layer. The Presentation Layer contains the User Interface Component, which handles interaction between users and the system. The Business Layer contains the User & Skill Management Component, Session Management Component, and Time Credit Management Component, which implement the core functionality of the platform. The Data Layer contains the Database Component, which is responsible for storing and retrieving system data.

The Layered Architecture was selected because it separates user interaction, business logic, and data management into distinct layers. This separation makes the system easier to understand, maintain, test, and modify. Changes made to one part of the system can also be handled with less impact on unrelated components.

2. Component Responsibilities

User Interface Component:

The User Interface Component belongs to the Presentation Layer and provides the interface through which users interact with the platform. It allows users to access skill-related features and perform actions such as booking skill-exchange sessions.

User & Skill Management Component:

The User & Skill Management Component belongs to the Business Layer. It is responsible for managing user-related information and the skills offered or requested by members of the platform.

Session Management Component:

The Session Management Component belongs to the Business Layer. It handles the process of booking and managing skill-exchange sessions between learners and mentors. It also manages session-related information required by other parts of the system.

Time Credit Management Component:

The Time Credit Management Component belongs to the Business Layer. It is responsible for

managing the time-credit system used by the platform. When a teaching session is successfully completed and verified, the corresponding time credits can be transferred between the learner and mentor.

Database Component:

The Database Component belongs to the Data Layer. It stores and retrieves persistent information required by the system, including user, skill, and session-related data.

3. Interface Justification

The components communicate through clearly defined interfaces so that each component can perform its own responsibility while interacting with other parts of the system when required.

User & Skill Interface:

This interface connects the User Interface Component with the User & Skill Management Component. It allows user interactions related to profiles and skills to be passed from the presentation layer to the appropriate business component.

Session Booking Interface:

This interface connects the User Interface Component with the Session Management Component. It is used for interactions related to booking and managing skill-exchange sessions. The provided and required interface notation represents the service relationship between these components.

Credit Transfer Interface:

This interface connects the Session Management Component with the Time Credit Management Component. It allows session-related events to trigger the appropriate time-credit operations, such as transferring credits after successful completion of a session.

User Data Interface:

This interface connects the User & Skill Management Component with the Database Component. It enables user and skill information to be stored and retrieved from the database.

Session Data Interface:

This interface connects the Session Management Component with the Database Component. It enables session-related information to be stored and retrieved as required by the system.

4. Design Rationale

The system was divided into separate components to achieve **separation of concerns**. Instead of placing all functionality inside a single component, responsibilities are divided among components according to their purpose. User and skill operations are handled separately from session management and time-credit management.

The User Interface Component is separated from the business components so that presentation logic does not become tightly coupled with the core functionality of the system.

Similarly, the Database Component is placed in a separate Data Layer so that data storage responsibilities remain separate from business logic.

The Session Management and Time Credit Management components are also separated because session handling and time-credit handling represent different responsibilities. The Credit Transfer Interface allows them to communicate when a completed session requires a credit operation without combining both responsibilities into one component.

5. Benefits of the Design

The proposed component design provides **modularity, maintainability, and scalability**. Each component has a clearly defined responsibility and communicates with other components through defined interfaces. This reduces unnecessary dependencies between different parts of the system.

The layered organization also makes future modifications easier. For example, the User Interface can be modified without redesigning the database, while changes to data storage can be handled primarily within the Data Layer. Similarly, individual business components can be modified or extended as new functionality is introduced.

Overall, the component design provides a clear and structured architecture for the Peer Skill Exchange & Mentorship Network while supporting the system's core functions of skill management, session management, and time-credit exchange.